COMPARATIVE ANALYSIS OF ARRAYS AND SINGLY LINKED LISTS FOR SORTING AND SEARCHING IN URBAN CARBON EMISSION DATA

(Working Duration: Monday of Week 6 to Monday of Week 10 – 30 Marks)

Background:

Cityville is a bustling urban area experiencing rapid growth in population and transportation. Every morning, thousands of residents commute to work, school, and other activities using cars, buses, bicycles, and by walking. While these daily movements are part of normal city life, they also contribute significantly to carbon emissions, affecting air quality and the environment. Concerned about rising pollution levels, the city council wants to encourage more sustainable commuting practices. To better understand residents' transportation habits and their impact on carbon emissions, a comprehensive survey was conducted across three areas of Cityville:

1. City A (Metropolitan City):
    - Adult commuters aged 18–60 traveling by cars, buses, bicycles, or walking.
2. City B (University Town):
    - Students and young adults aged 6–25 commuting by bicycles, walking, school buses, or carpools.
3. City C (Suburban/Rural Area):
    - Residents of all ages using cars, local buses, and bicycles.

The survey collected detailed data for each resident, including:

1. Resident ID – a unique identifier for each participant.
2. Age – the age of the resident.
3. Mode of Transport – type of transport used for commuting (Car, Bus, Bicycle, Walking, School Bus, Carpool).
4. Daily Distance (km) – the distance traveled each day by the resident.
5. Carbon Emission Factor (kg CO_2/km) – the estimated carbon emissions per kilometer for the chosen mode of transport.
6. Average Day per Month – the average number of days per month that the resident uses the specified mode of transport.

City planners now aim to analyse this data to identify which transport modes contribute most to carbon emissions, compare emissions across cities, and design policies that encourage greener commuting habits.

To perform this analysis efficiently, the city's research team decided to store and process the data using arrays and singly linked lists, and to apply various sorting and searching algorithms. By comparing the performance of these data structures and algorithms, the council can determine the most effective methods to organize and analyse large urban datasets, while gaining valuable insights that support environmentally friendly transportation initiatives.

Assignment Tasks

1. This assignment must be completed in teams, with a minimum of **THREE (3)** members and a maximum of **FIVE (5)** members per team.

2. **Datasets Provided**
   a. Three datasets are provided for this assignment:
      - Dataset 1 – City A **(Metropolitan City):** Adult commuters traveling by cars, buses, or bicycles, or walking.
      - Dataset 2 – City B **(University Town):** Students and young adults commuting by bicycles, walking, school buses, or carpools.
      - Dataset 3 – City C: **(Suburban/Rural Area):** Residents of all ages using cars, local buses, and bicycles.

3. **Program Implementation**
   a. Implement Dataset 1, Dataset 2, and Dataset 3 using **arrays.**
   b. Implement the same datasets using **singly linked lists.**
   c. Ensure the data structures allow easy access for **sorting, searching, and analysis operations**. (each person one)
   d. Therefore, this assignment requires the development of **two (2) separate programs**:
      - One focusing on array implementation
      - The other on linked list implementation (compare)

4. **Age Group Categorization**
   a. Recategorize residents into the following age groups:
      - 6–17: Children & Teenagers
      - 18–25: University Students / Young Adults
      - 26–45: Working Adults (Early Career)
      - 46–60: Working Adults (Late Career)
      - 61–100: Senior Citizens / Retirees

   b. For each age group, determine the below:
      - Most preferred **mode of transport**.
      - **Total carbon emissions** produced.
      - **Average carbon emission per resident**.

$$\text{Average Emission} = \frac{\text{Total Emission in Age Group}}{\text{Number of Residents in Age Group}}$$

- Lab : individual
- Doc : group
- Presentation : 1 video, 1 live

5. Carbon Emission Analysis

a. Calculate **total carbon emissions per dataset**.
b. Determine **carbon emissions per mode of transport**.
c. Compare **emissions across datasets and age groups**.
d. Display results as **text-based tables** in the console.
Sample:

```
Age Group: 18-25 (University Students)
--------------------------------------
Mode of Transport       Count    Total Emission (kg CO2)    Average per Resident
Bicycle                 50       0                          0
Walking                 30       0                          0
School Bus              25       120                        4.8
Carpool                 20       60                         3
--------------------------------------
Total Emission for Age Group: 180 kg CO2
```

6. Sorting Experiments

a. Sort datasets by **Age**, **Daily Distance**, or **Carbon Emission**.
b. Implement **at least one (1) sorting algorithms**.
c. Compare **time complexity and memory usage** between arrays and singly linked lists.
d. Display results as **text-based tables** in the console. (at least 3 tables)

7. Searching Experiments

a. Search for residents matching specific criteria, such as:
   - Age group (e.g., 26–45)
   - Mode of transport (e.g., Car or Bicycle)
   - Daily distance threshold (e.g., >15 km)
b. Implement **different search algorithms** on unsorted / sorted data.
c. Compare **time complexity and memory usage** for arrays and singly linked lists.
d. Display search results as **text tables in the console**.

8. Performance Analysis

a. Record **execution time and memory usage** for all sorting and searching experiments
b. Analyse under which conditions **arrays or singly linked lists perform better**.
c. Discuss any **advantages or disadvantages** of each data structure in real-world data processing.

9. Insights and Recommendations

a. Compare **carbon emissions and mode preferences** across datasets and age groups using **text-based tables.**
b. Identify which **age groups contribute most to emissions,** and which prefer bicycles or cars.
c. Suggest **practical recommendations** for city planners to encourage sustainable commuting based on age-specific patterns. (future plan)

AI USAGE POLICY FOR CODING ASSIGNMENTS

1. **Students are strictly prohibited** from using artificial intelligence (AI) tools to generate complete or partial solutions, source code, algorithms, or program logic for this assignment.

2. The use of AI tools is permitted **only for supportive purposes**, such as:
   - Understanding programming concepts or syntax
   - Identifying relevant keywords or technical terms
   - Clarifying error messages or documentation terminology

3. **All submitted code must be the student's own original work**. Lecturers reserve the right to conduct oral questioning, code tracing, or live demonstrations to verify the authenticity and understanding of submitted work.

4. Therefore, students must be able to *explain and justify every part of their code* during evaluation or questioning.

5. If AI tools are **used for assistance,** students *must clearly declare the purpose of AI usage* e.g., "AI was used to identify appropriate keywords related to file handling in Python"

6. **Copying, paraphrasing, or adapting AI-generated code or solutions**, whether modified or unmodified, is **considered a violation of academic integrity.**

7. **Any misuse of AI tools**, including generating solutions beyond permitted assistance, **may result in penalties** including:
   - **Zero (0)** marks for the assignment **OR**
   - **Disciplinary action** in accordance with institutional policies

Submission Guidelines #1: Program and Video Submission – Lab Work #1 (15 Marks)

1. Your team is required to utilize C++ programming to develop this prototype, comprising two programs in this section.
   - *Do not use GUI, charts, or external visualization libraries.*
   - *Organize data neatly in text tables using formatting (e.g., fixed-width columns, separators).*

2. **Built-in containers such as <list>, <vector>, etc. are not allowed** in this assignment. All containers need to be self-created.

   Refer to the link: https://www.geeksforgeeks.org/containers-cpp-stl/ for further information on built-in containers in STL C++.

3. The program will be enhanced if your team effectively implements and compares various useful searching or sorting algorithms in your developed prototype.

4. The evaluation criteria for this lab work #1 include assessing the clarity and structural design of the code, as well as the quality of comments and adherence to good programming practices. (e.g., indentation, meaningful identifier names, comments, etc.).

5. **Creativity is encouraged.**
   The best solution is not limited to the core features specified in this assignment. Teams are encouraged to extend the system with additional logical functionalities, provided that all enhancements are clearly related to and supported by data structure concepts.

6. **Plagiarism Policy and Penalty.**
   Plagiarism in any form, including copying code, algorithms, documentation, or analysis from other groups, seniors, online sources, or AI-generated content without proper acknowledgment, is strictly prohibited.

   *Any group found to have committed plagiarism will receive a **FAIL (0 marks) for the assignment**. Further disciplinary action may be taken in accordance with the institution's academic integrity policy.*

   All submitted work must be original and developed solely by the assigned group members.

7. Team leader must submit the system solution in a ZIP file format to Moodle by Monday of Week 10, no later than 5:00 pm.

   - use the lec give one (X use excel file to sort, X file in font)

   a) Include only the .cpp, .hpp, and .csv/text files in the ZIP file.
   b) The ZIP file must follow this naming format:

> *"<GroupNo>_<Team Leader ID >_<Member 1 ID >_<Member 2 ID >_<Member 3 ID >.zip"*
>
> For example, **"G1_TP012345_TP014556_TP067554_TP034325.zip"**

8. Each team leader must also upload **<u>ONE (1)</u>** video recording to the Moodle system by Monday of Week 10, no later than 5:00 pm.

    a) The maximum duration for the overall video recording is 30 minutes.
    b) All members must participate in the video recording, presenting and explaining the development process in detail, with each member speaking for no more than 5 minutes.
    c) Each member must relate their explanation to the workload matrix distribution table provided in the Word document. (add color remark for [illegible])
    d) Compress the final video recording to under 200MB before submitting.
    e) If the video exceeds the specified time limit (30 minutes), it will only be assessed up to the specified duration.
    f) Videos must be recorded at normal speed (1x) and cannot be sped up or adjusted to meet the demo video duration requirements.
    g) The video recording file must adhere to the following name format:

“<GroupNo>_<Team Leader ID >_<Member 1 ID >_<Member 2 ID >_<Member 3 ID >.mp4”

For example, **“G1_TP012345_TP014556_TP067554_TP034325.mp4”**

9. Refer to the **<u>Page 6</u>** for marking criteria of this Lab Evaluation Work #1 submission.

Submission Guidelines #2: Documentation Submission – Solution Work (15 Marks)

1. Create a **Word document** (refer Appendix 1 in page 7) that explains the overall efficiency and limitations of your developed system

2. Clearly demonstrate the differences in implementation and performance between arrays and linked lists based on the ***operations performed*** in this specific analysis scenario.

3. Present results that clearly show the advantages and disadvantages of using arrays and linked lists in this specific sentiment analysis scenario.

4. Include a summary of all your hypotheses, observations, experimental results, justifications, and reflections, etc. in your word documentation.

5. Your team must include a workload distribution matrix table after the cover page, showing each member's contributions to the project.

   Note that this table will impact each member's personal final mark in Lab Evaluation Work #1 based on their stated contribution percentage.

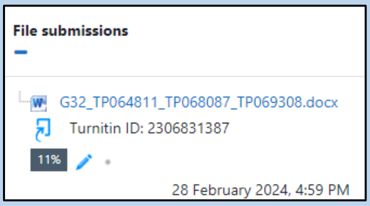
6. If you use some code which has been taken or adapted from another source (book, magazine, internet, forum, etc.) then this must be **cited and referenced** using **APA Referencing Style within your source code**, and this must be mentioned explicitly in the **report**. Failure to reference code properly will be treated as plagiarism.

7. The team leader is required to upload a word document before / on Monday of Week 10, no later than 5:00 pm.

   The documentation should be named using format:

   “<GroupNo>_<Team Leader ID >_<Member 1 ID >_<Member 2 ID >_<Member 3 ID >.docx”

   For example, **“G1_TP012345_TP014556_TP067554_TP034325.docx”**

10. Refer to the **Page 6** for marking criteria of this Solution Work submission.

MARKING CRITERIA

(Lab Evaluation Work #1 - 15 MARKS)

This Lab Evaluation Work #1 will be evaluated according to the following performance criteria:

Assessment Components	Inclusive	15 Marks
CLO3: Lab Evaluation Task #1 – 30-Minute Video Recording ***(Assessment will be based on individual performance)***		
Practical Skills: Use of Data Structures & Algorithms + Personal Understanding		
Utilization of data structures	Technical Proficiency	
Implementation of relevant algorithms	Technical Proficiency	
Demonstrates understanding of data structures/algorithms used	Comprehensive Understanding	
Justifies choices of structures/algorithms	Insightful Justification	

MARKING CRITERIA

(Solution Work - 15 MARKS)

This solution work will be evaluated according to the following performance criteria:

Assessment Components	Inclusive	15 Marks
CLO2: Solution Work - Documentation ***(Assessment will be based on group component)***		
Theoretical Explanation (e.g., Data Structures, Algorithms)	Clear explanation	
Input Output Screenshots	Adequate Screenshots	
Summary Discussions (Inclusive Time and Space Complexity)	Clear and insightful analysis	
Conclusion & Reflection (Other Relevant / Importance Discussions)	Clearly highlighted and insightful	
Content Organization	Well-structured and logical flow	

Approximation of Total Pages for the documentation: **40 (max).**
Approximation of Words for the documentation: **6000 words (max)**

APPENDIX 1:
SOLUTION WORK - DOCUMENTATION
(15 MARKS)

The report outline as below:

- **Cover Page** (1 page)
  - Module Code and Name
  - Intake Code
  - Group Number
  - Member List.

- **Workload Matrix Table with signature** (1 page)

- **Theoretical Explanation** (2 - 4 pages)
  - Elaborate on the data manipulations in chosen data structures.

- **Input Output Screenshots** (5 - 15 pages)
  - System input output screenshots

- **Summary Discussions** (5 - 15 pages)
  - System efficiency (*including actual execution time, time efficiency, space efficiency etc.*).
  - Summarize and discuss all the observations / thoughts made during the development.
  - Critically summarize the strengths and weaknesses of the code you developed.

- **Conclusion and Reflection** (1 - 2 pages)
  - Summary of your works
  - Describe your future work in terms of your system weaknesses, if any.
  - Additional personal thoughts on this assignment, if any.

- **References (if applicable)** (1 page)
- **Appendix (if applicable)** (1 page)

Approximation of Total Pages	: **40 (max).**
Approximation of Words	: **6000 words (max)**